

4. COURSE OBJECTIVES OF THE COURSE

Week No.	Topics	Required Reading	Key Date
1	Introduction Introduction to Artificial Intelligence & Machine Learning. Types of Machine Learning (<i>supervised, Unsupervised, Semi Supervised, Reinforcement Machine Learning</i>), Applications of ML Programming for ML (Python, R). Introduction of Libraries (scikit-learn, etc). Intro to Dataset in ML Training & Testing Dataset. Explore the Dataset. Use of Kaggle.	C.M Chapter #1 Page#1-11 OT Page# 11	

Introduction to Artificial Intelligence and Machine Learning

I. What is Artificial Intelligence (AI)?

- ❑ AI is a field of computer science that aims to create machines capable of performing tasks that typically require human intelligence.
- ❑ AI systems can learn from data, reason, solve problems, understand natural language, and perceive their environment.



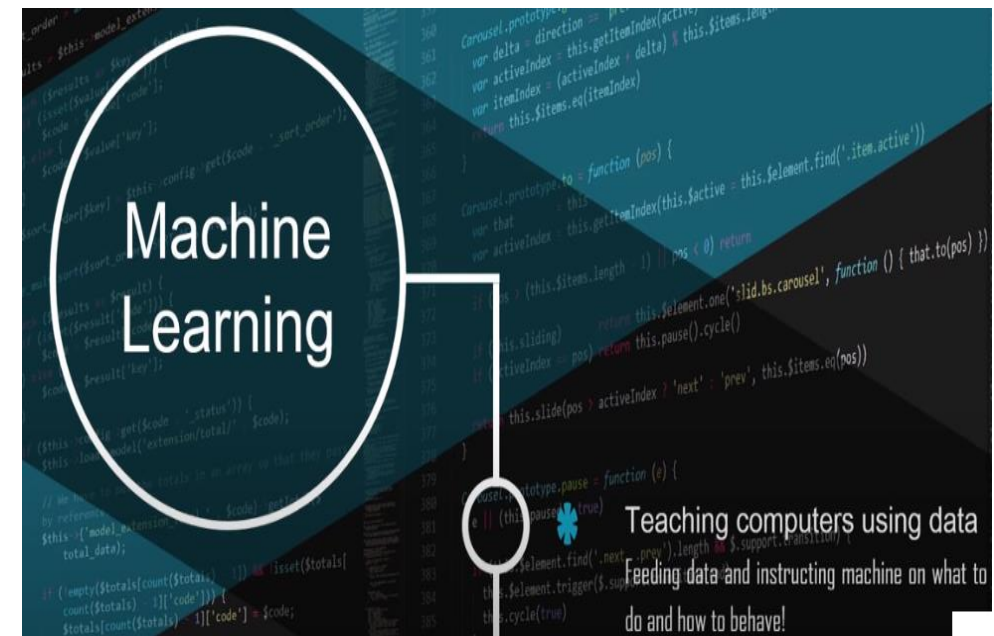
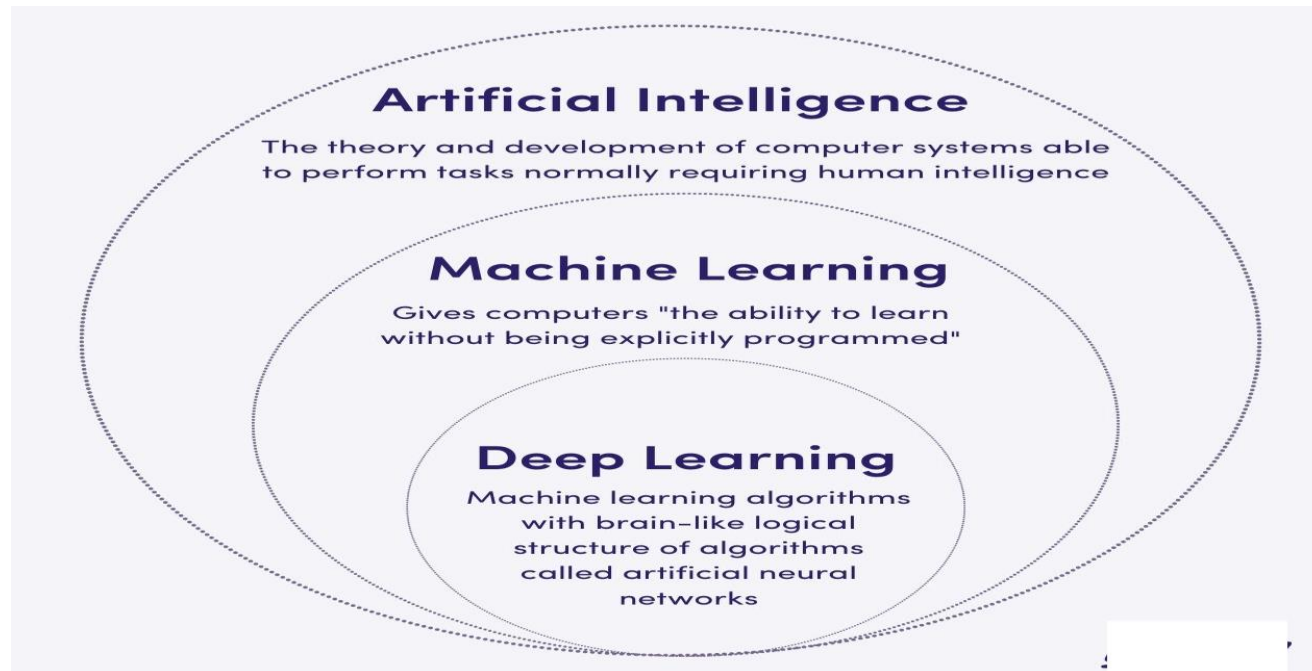
Introduction to Artificial Intelligence and Machine Learning

II. Machine Learning: A Subset of AI

- ❑ Machine Learning is a subset of AI that focuses on creating algorithms and models that can learn from data.
- ❑ Instead of explicitly programming tasks, ML systems improve their performance through experience.

III. The Importance of AI and ML

- ❑ AI and ML have profound applications in various domains, including healthcare, finance, robotics, and more.
- ❑ They enable automation, data-driven decision-making, and the development of intelligent systems.



1989

Yann LeCun, Yoshua Bengio and Patrick Haffner demonstrate how convolutional neural networks (CNNs) can be used to recognize handwritten characters, showing that neural networks could be applied to real-world problems.

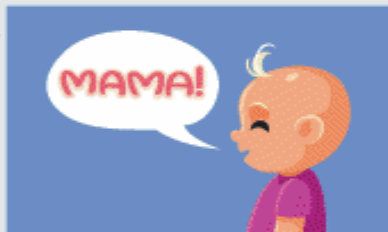


a model-free reinforcement algorithm that seeks the best action to take in any current state.

Axcelis releases Evolver, the first commercially available genetic algorithm software package for personal computers.

1997

Sepp Hochreiter and Jürgen Schmidhuber propose the Long Short-Term Memory recurring neural network, which could process entire sequences of data, like speech or video.



1992

Gerald Tesauro invents a program capable of playing backgammon called TD-Gammon, based on an ANN and rivaling top-tier backgammon players.

IBM's **Deep Blue** defeats **Garry Kasparov** in a historic chess rematch, the first defeat of a reigning world chess champion by a computer under tournament conditions.



1998

A team led by Yann LeCun releases a data set known as the MNIST (Modified National Institute of Standards and Technology) database, which becomes widely adopted as a handwriting recognition evaluation benchmark.

2000

University of Montreal researchers publish "A Neural Probabilistic Language Model," which suggests a method to model language using feed-forward neural networks.



2002

The first open source machine learning library, Torch, is released, providing interfaces to deep learning algorithms implemented in C.

2006

Netflix launches the Netflix Prize competition with the goal of creating a machine learning algorithm more accurate than Netflix's proprietary user recommendation software.

2006

Psychologist and computer scientist **Geoffrey Hinton** coins the term *deep learning* to describe algorithms that help computers recognize different types of objects and text characters in pictures and videos.

IBM Watson originates with the initial goal of beating a human on the iconic quiz show *Jeopardy!*. In 2011, the question-answering computer system defeats the show's all-time (human) champion **Ken Jennings**.



2010

Fei-Fei Li starts working on the ImageNet visual database (introduced in 2009), which becomes a catalyst for the AI boom and the basis of an annual competition for image recognition algorithms.

Introduction to Artificial Intelligence and Machine Learning

IV. Key Concepts in Machine Learning

- ❑ **Data**: ML algorithms learn from data, which can be structured or unstructured.
- ❑ **Training**: ML models are trained on historical data to make predictions or decisions.
- ❑ **Types of Learning**: ML includes supervised, unsupervised, and reinforcement learning.
- ❑ **Applications**: ML is used for tasks like image recognition, natural language processing, and recommendation systems.

V. Real-World Examples of AI and ML

- ❑ **Chatbots**: AI-driven virtual assistants that understand and respond to natural language queries.
- ❑ **Self-Driving Cars**: Autonomous vehicles that use ML to navigate and make driving decisions.
- ❑ **Medical Diagnosis**: ML models can analyze medical data to aid in disease diagnosis.

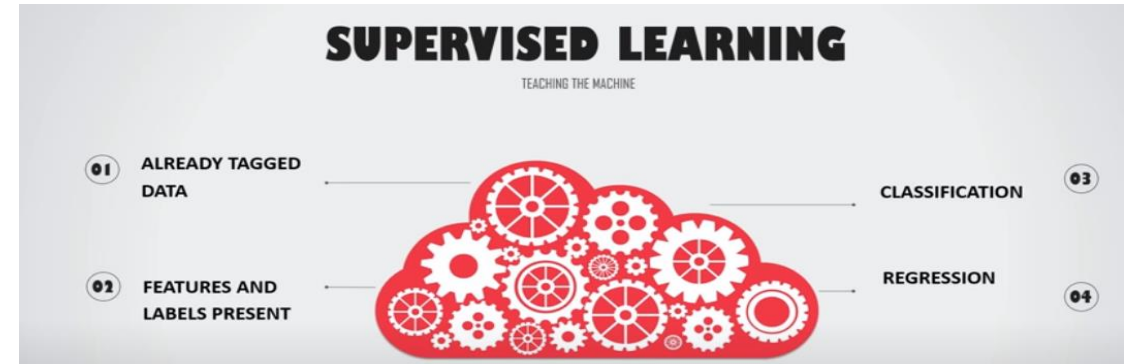
Introduction to Artificial Intelligence and Machine Learning

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What is Supervised Learning?

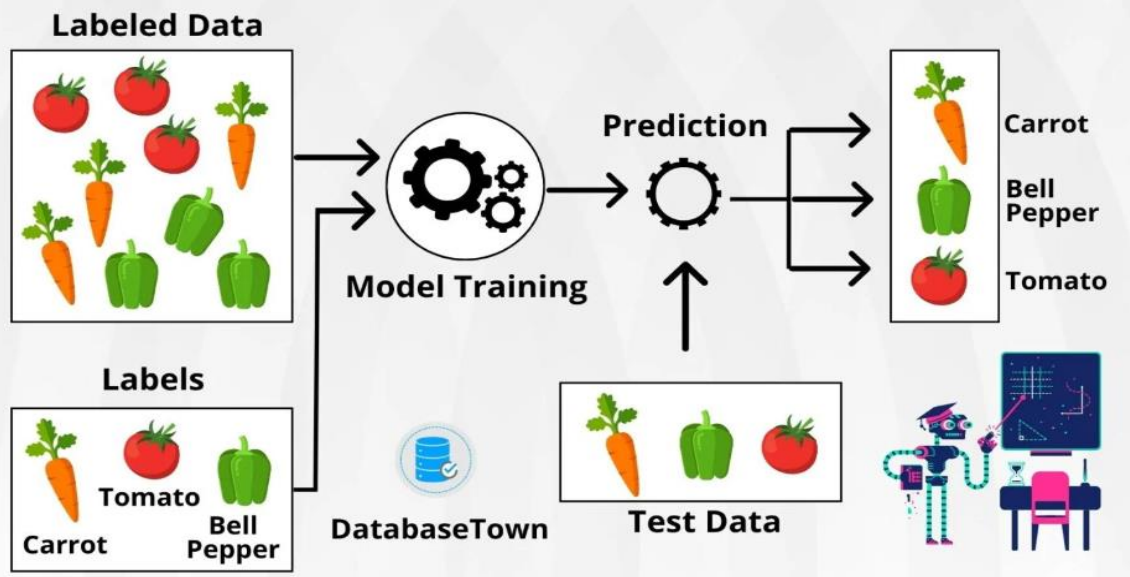
Machine knows something beforehand



What is Unsupervised Learning?

All machine knows is the data in front of it. No Features! No Labels





SUPERVISED LEARNING

MACHINE KNOWS THE FEATURES



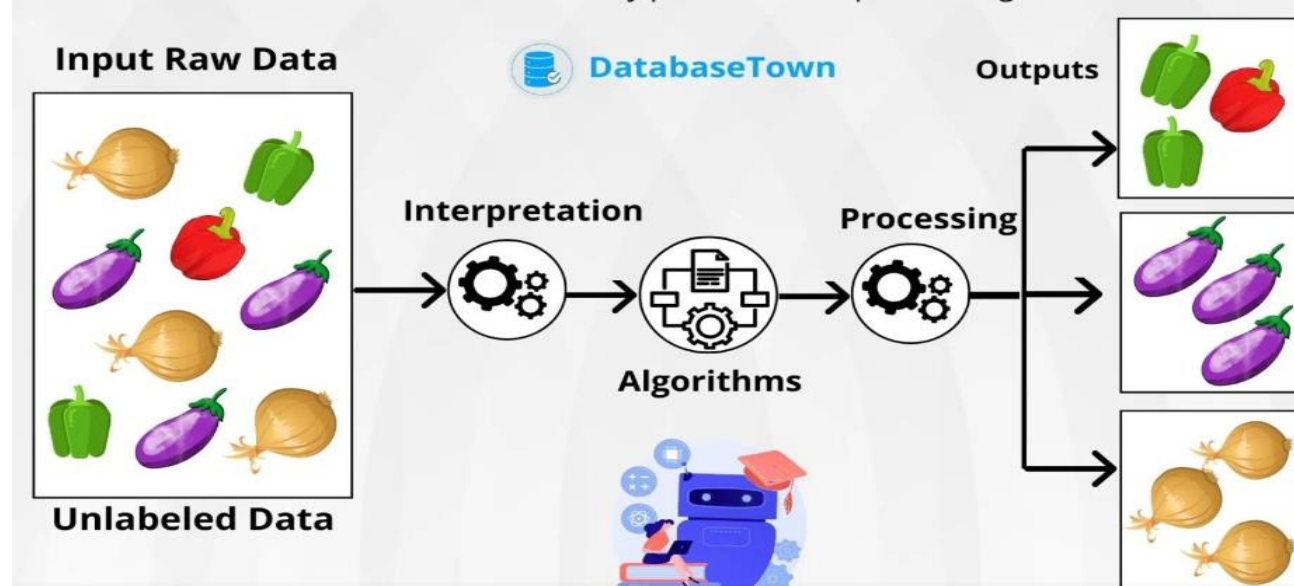
Classification

Classify which label a given set of features belong to!



Regression

Find out the value of the label using previous data!



UNSUPERVISED LEARNING

MACHINE SEEING THE DATA FOR THE FIRST TIME



Clustering

Discover the inherent groupings in the data, such as grouping customers by purchasing behavior.



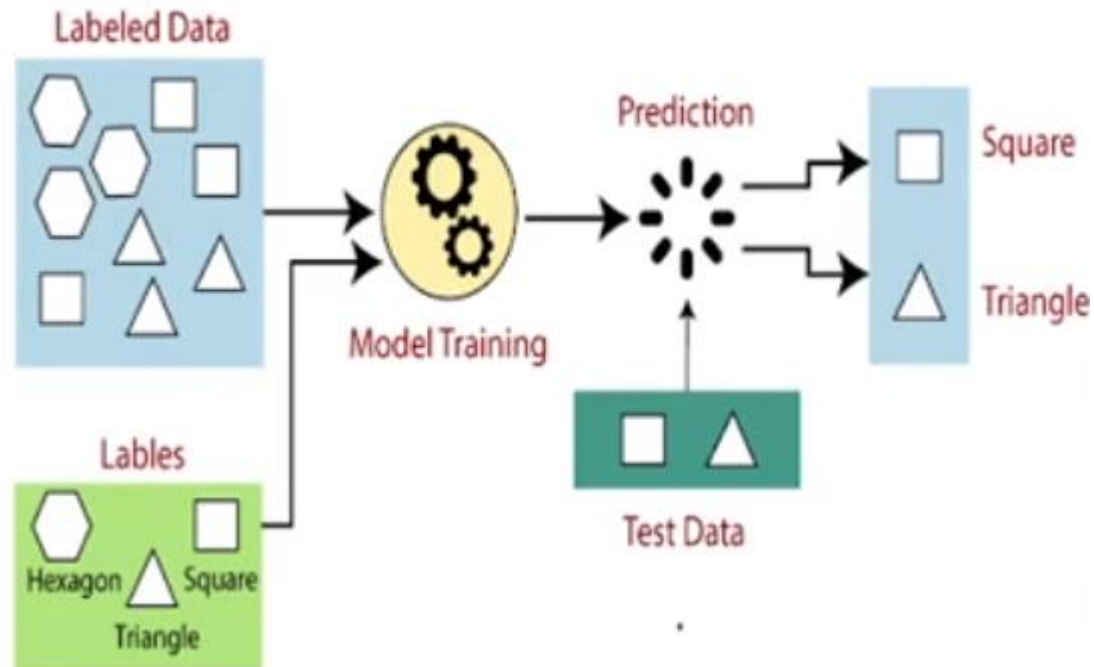
Association

Association rule learning problem. such as people that buy X also tend to buy Y

Reinforcement Learning

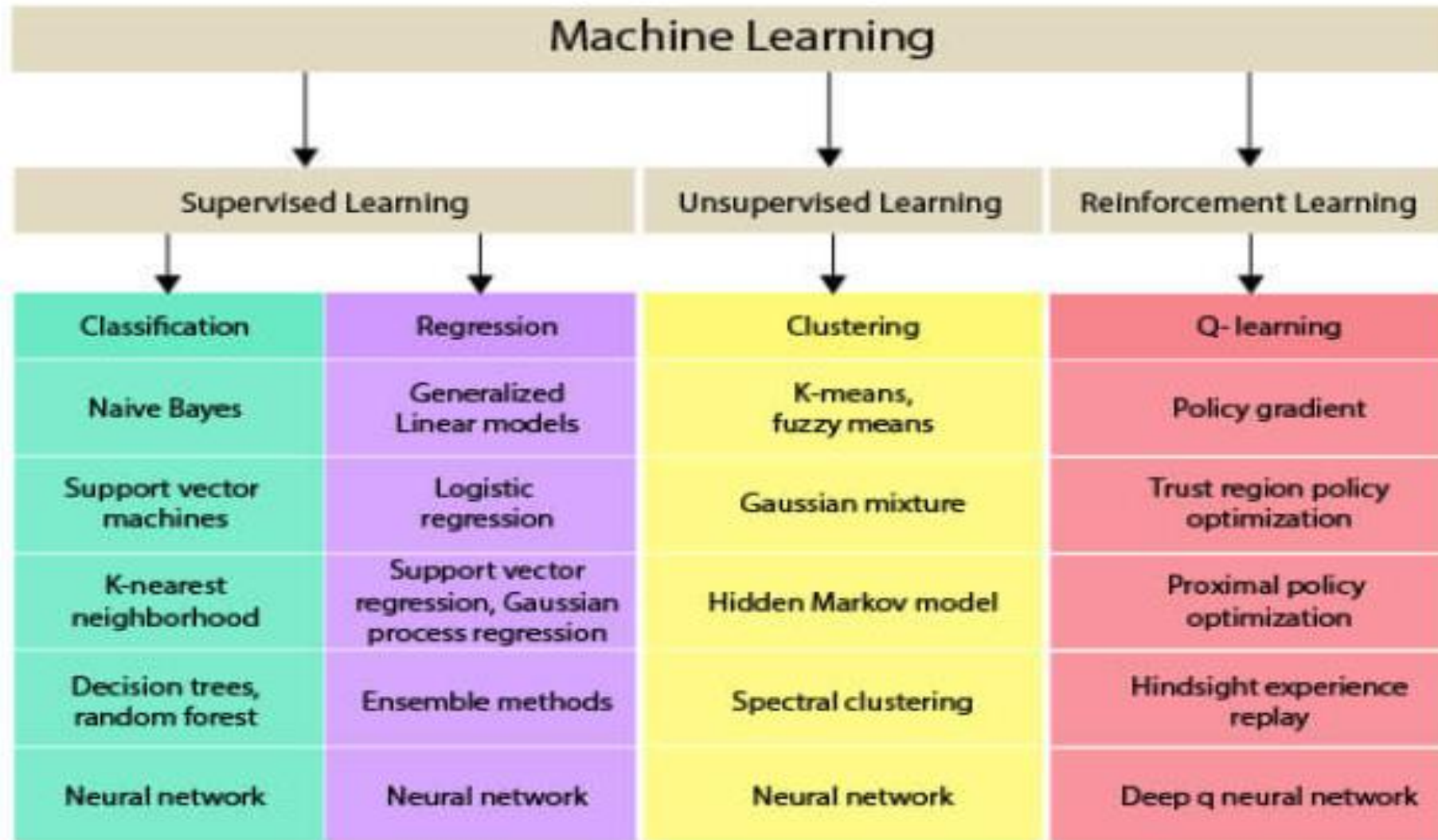


Supervised Learning



Logistic regression,SVM,KNN,Naïve Bayes,Decision Trees

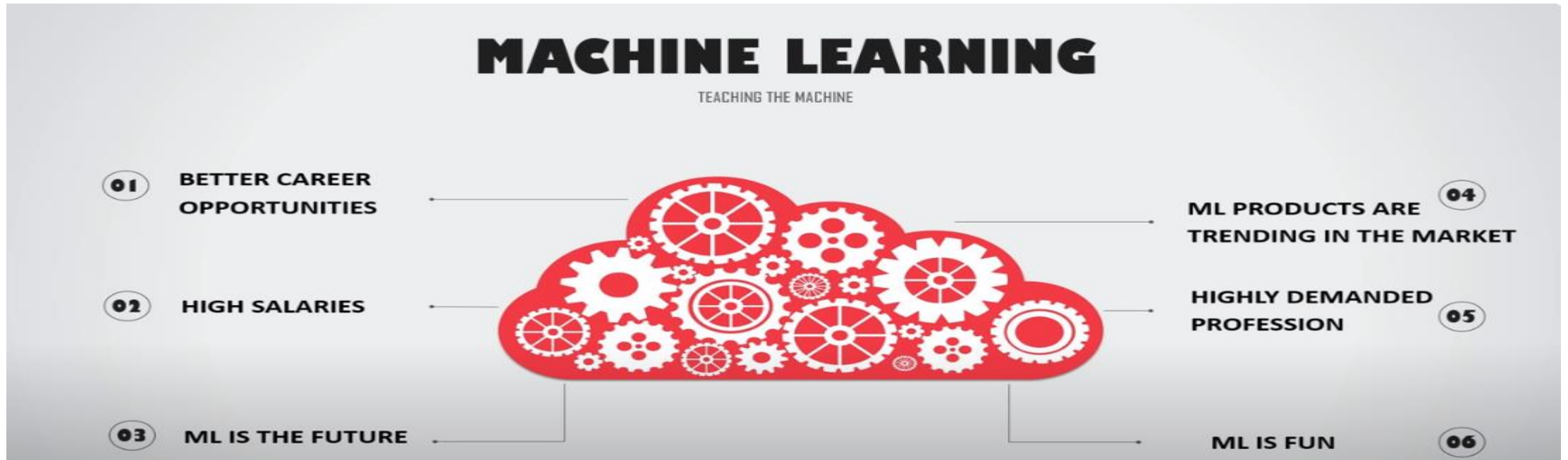
Machine Learning Algorithms



Introduction to Artificial Intelligence and Machine Learning

VI. The Future of AI and ML

- ❑ AI and ML continue to advance, with deep learning and neural networks at the forefront.
- ❑ Ethical considerations, privacy, and AI safety are important topics in AI development.
- ❑ The job market for AI and ML professionals is growing rapidly.



Why Programming of Machine Learning in Python?

Programming Machine Learning in Python is a popular choice for several reasons:

1. Ease of Learning:
2. Abundant Libraries and Frameworks: :
3. Community and Support:
4. Flexibility:
5. Libraries for Data Manipulation:
6. Visualization Tools:
7. Open Source:
8. Interoperability:

Why Python
For Machine Learning

01
Python Is Easy

02
Python Is Fast Enough

03
Python Has ML Libraries

04
Python Is Cross Platform

The infographic features the Python logo at the bottom of each of the four numbered sections. The logo is a stylized 'P' composed of blue and yellow segments.

Introduction of Libraries of Python



Introduction of Libraries of Python

NumPy (Numerical Python):

Purpose: **NumPy is fundamental for numerical and array operations.** It provides support for multi-dimensional arrays and matrices, along with mathematical functions to operate on them efficiently.

Key Features: Arrays, element-wise operations, linear algebra, random number generation.

Example: Calculating the mean and standard deviation of a dataset.

Pandas:

Purpose: **Pandas is essential for data manipulation and analysis.** It offers data structures like DataFrames and Series, making it easier to work with structured data.

Key Features: DataFrames, data cleaning, data aggregation, time-series analysis.

Example: Loading, filtering, and summarizing a dataset.

Matplotlib:

Purpose: **Matplotlib is a powerful library for creating static, animated, or interactive visualizations in Python.**

Key Features: Line plots, scatter plots, bar charts, histograms.

Example: Plotting a graph to visualize data trends.

Introduction of Libraries of Python

Seaborn:

Purpose: Seaborn is built on top of Matplotlib and provides a higher-level interface for creating informative and attractive statistical graphics.

Key Features: Enhanced data visualization, support for complex plots.

Example: Creating a heatmap or pair plot for data analysis.

Scikit-Learn:

Purpose: Scikit-Learn is a machine learning library that offers tools for data preprocessing, model selection, evaluation, and deployment.

Key Features: Classification, regression, clustering, model evaluation, feature selection.

Example: Building a machine learning model for predicting house prices.

TensorFlow and PyTorch:

Purpose: TensorFlow and PyTorch are deep learning frameworks used for building and training neural networks. They are widely employed in applications like computer vision and natural language processing.

Key Features: Deep learning, neural network design, GPU acceleration.

Example: Creating a convolutional neural network (CNN) for image recognition

Introduction of Libraries of Python

Flask and Django:

Purpose: **Flask and Django are web development frameworks.** Flask is lightweight and ideal for small to medium-sized web applications, while Django is a comprehensive framework for larger projects.

Key Features: Routing, templating, database integration (Django includes an ORM).

Example: Building a simple web app to collect user feedback.

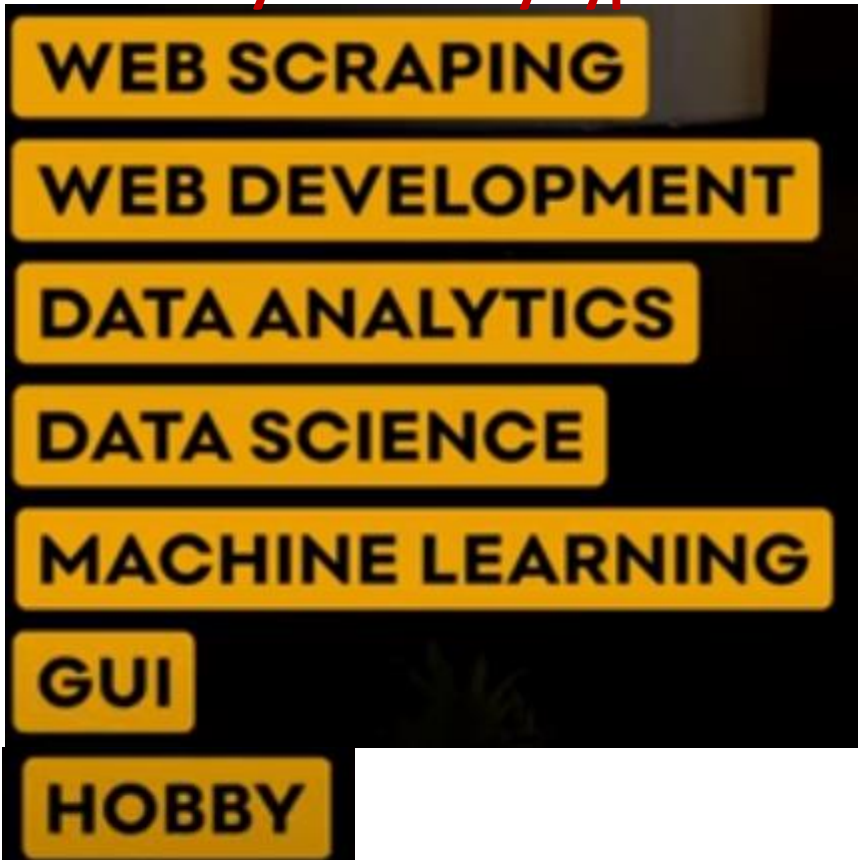
Requests:

Purpose: **The Requests library simplifies HTTP requests, making it easy to interact with web services and APIs.**

Key Features: HTTP GET, POST, PUT, and DELETE requests, handling response data.

Example: Sending a GET request to retrieve data from a RESTful API.

Summary of Library types



Request Library

Web Scraping

Web Development

Beautiful Soup Library

Data Scraping

We Can Get HTML

Selenium Library

To Automate Web Browser

We Can Scrap The Data

Django Library

Used For Web Development

Most Used Library

Flask Library

Used For Web Development

Build Websites From Scratch

NumPy Library

Used For Data Science

Used For Scientific Calculation

Pandas Library

Used For Data Manipulation

Dealing With Missing Data

Matplotlib Library

Used For Creating Charts

Keras Library

Used For Deep Learning Models

Kivy Library

Used For Creating GUI Applications

We Can Create Cross-Platform Application

Pygame Library

Used For Game Development

NLTK Library

Used For Data Processing

Used For Data Organization

TensorFlow Library

Also A Deep Learning Library

Scikit Library

It's Provides Us All Machine Learning Algorithms